



Large Language Models Enhanced Personalized Graph Neural Architecture Search in Federated Learning

汇报人： 憨芳静

时间： 2026.04.09



- **Background**
- **GNAS(2020)**
- **PFGNAS(2025)**
- **Results**
- **Future Work**

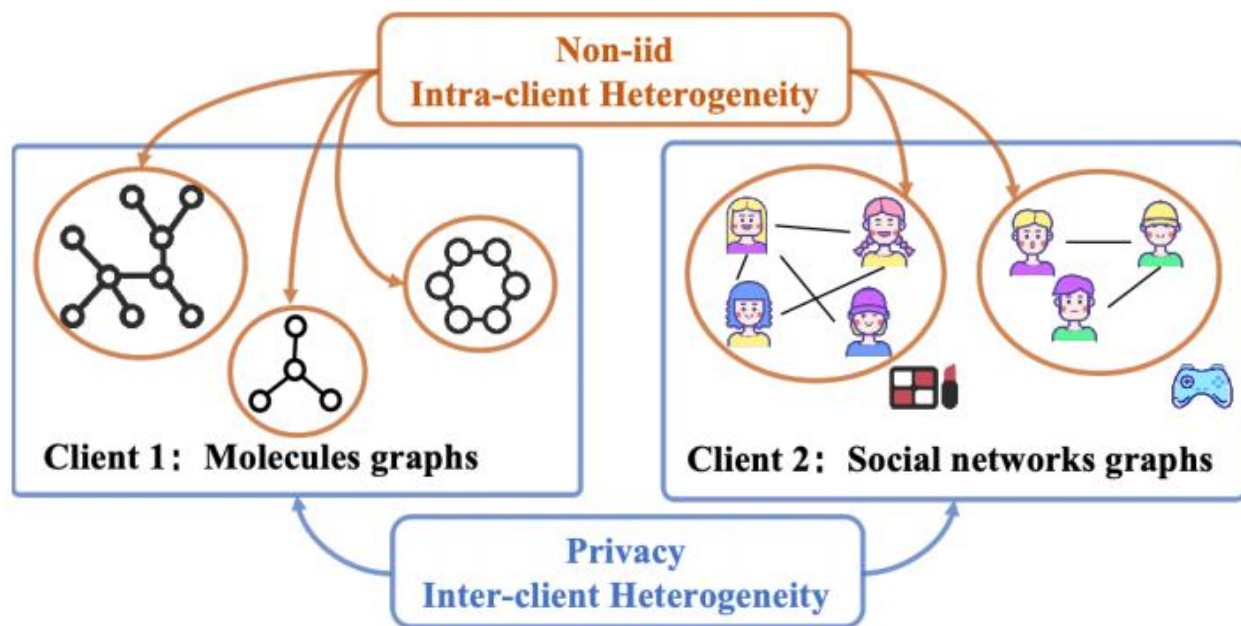


Figure 1: Heterogeneity of the graph data between and within clients. Client 1 exhibits significant structural heterogeneity, while Client 2 demonstrates notable feature heterogeneity.

The Vision (IID Expectation)

Break cross-institutional data silos.

Train together.

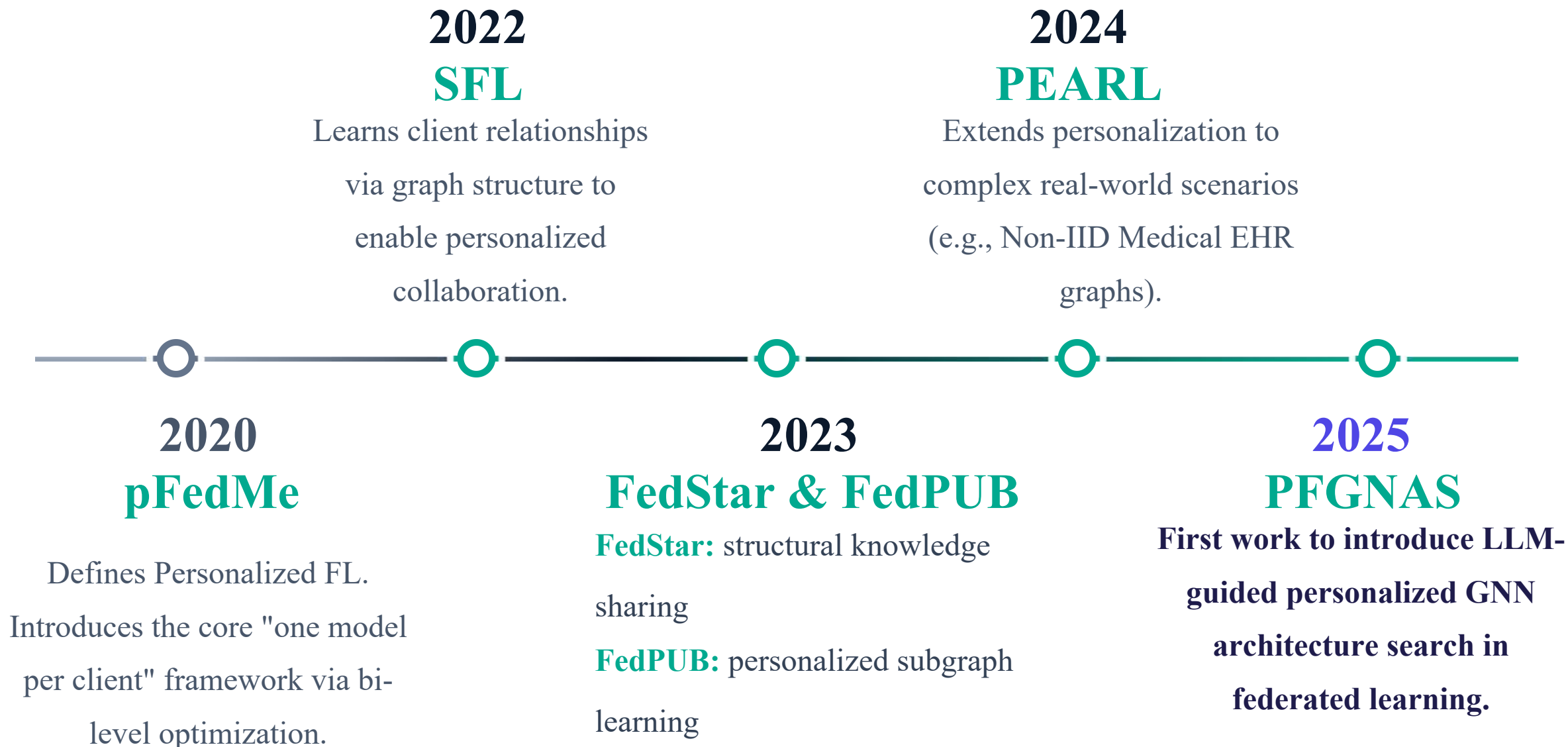
Ensure privacy .

The Reality (Non-IID)

Highly heterogeneous graph data:

Massive topological differences &
heavily skewed feature distributions.

Consequence: Client Drift



Structure-Level

Utilizes manual decoupling strategies. The limitation is a high reliance on expert experience and extreme sensitivity to hyperparameters.

Parameter-Level

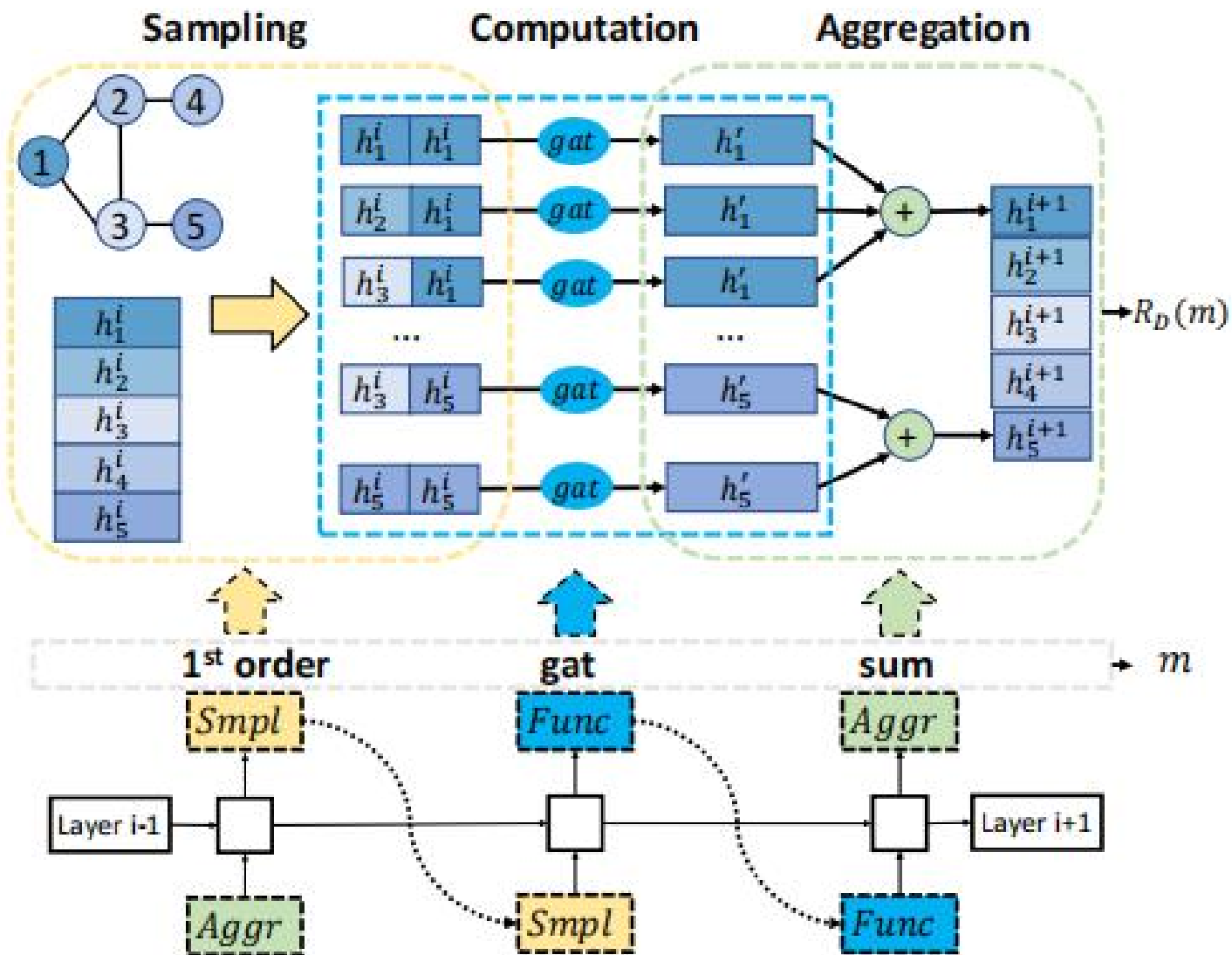
Relies on regularization constraints and meta-learning. The core limitation is that the architecture remains strictly homogeneous across all clients.

GNAS Integration

Automated, data-driven architecture search designed to automatically tailor and fit GNN structures individually per client.



GNAS & The LLM Era



$$m^* = \arg \max_{m \in \mathcal{M}} \mathbb{E}[R_D(m)].$$

Controller : RNN

An architecture : m

$RD(m)$: reward

Operators	Values
<i>Smpl</i>	<i>first_order</i>
<i>Func</i>	$e_{uv}h_u$
<i>Aggr</i>	<i>sum, mean, max, mlp</i>
<i>Read</i>	<i>avg</i> , for the last layer <i>concat</i> , otherwise
activate function σ	<i>sigmoid, tanh, relu, identity,</i> <i>softplus, leaky_relu, relu6, elu</i>
multi-head k	1, 2, 4, 6, 8, 16
output dimension d	8, 16, 32, 64, 128, 256, 512

Table 1: Operators of search space $\mathcal{M}$

1. Centralized Assumption

Requires full access to global graph data

Not applicable in federated environments

2. Lack of Personalization

Searches for a single global architecture

Fails under heterogeneous client distributions

3. High Computational Cost

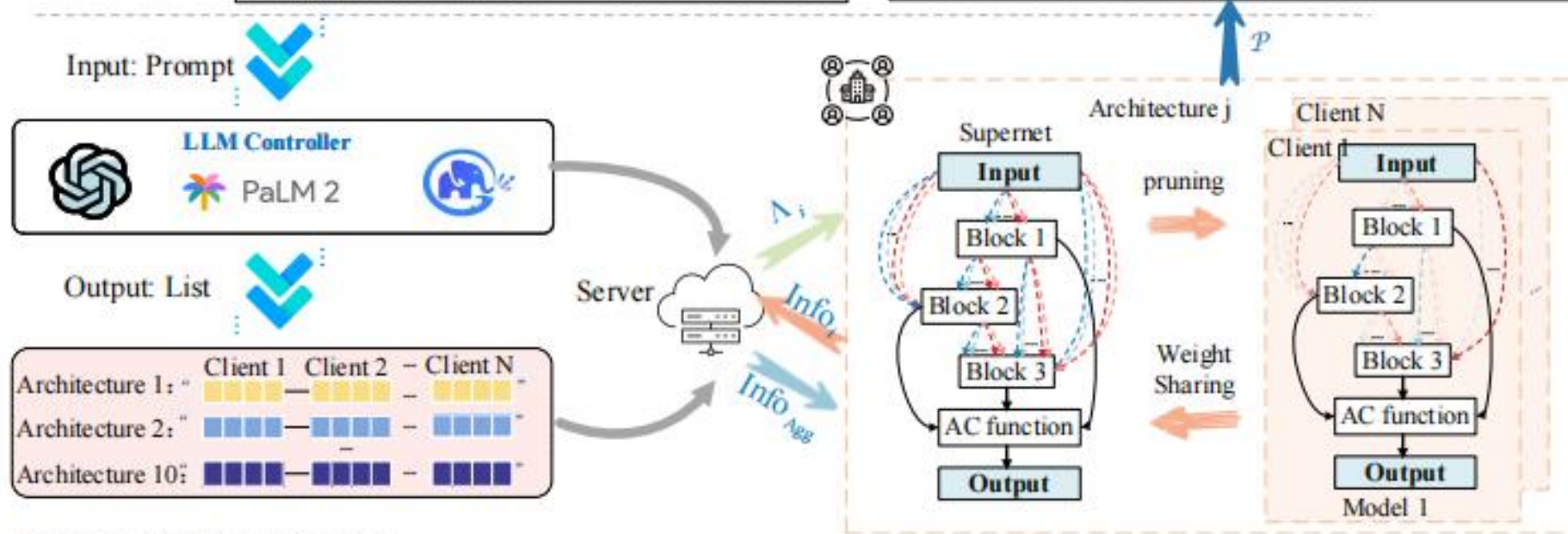
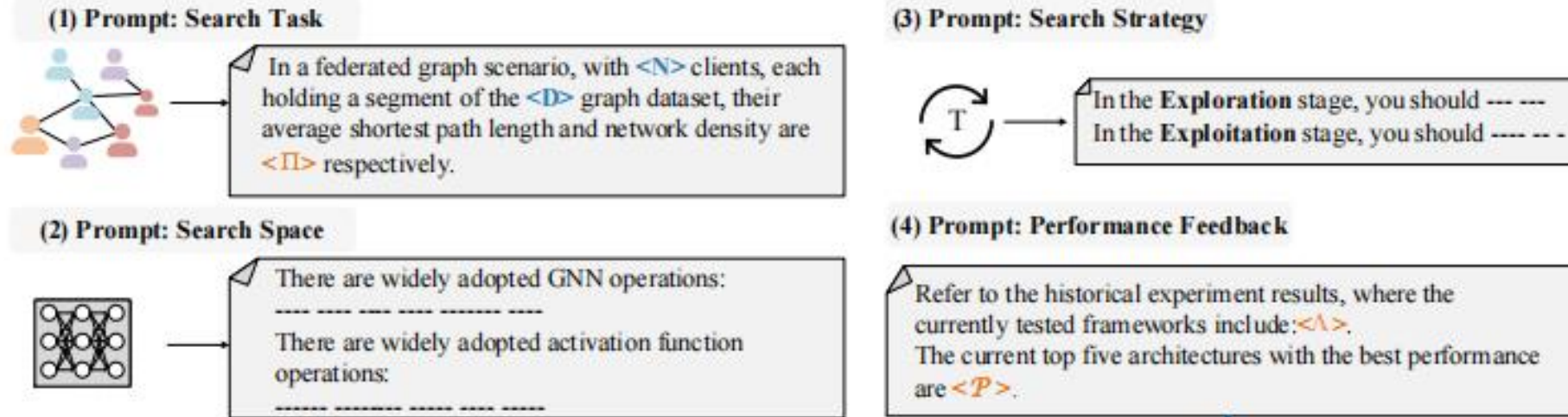
Requires repeated architecture evaluation

Inefficient under communication constraints



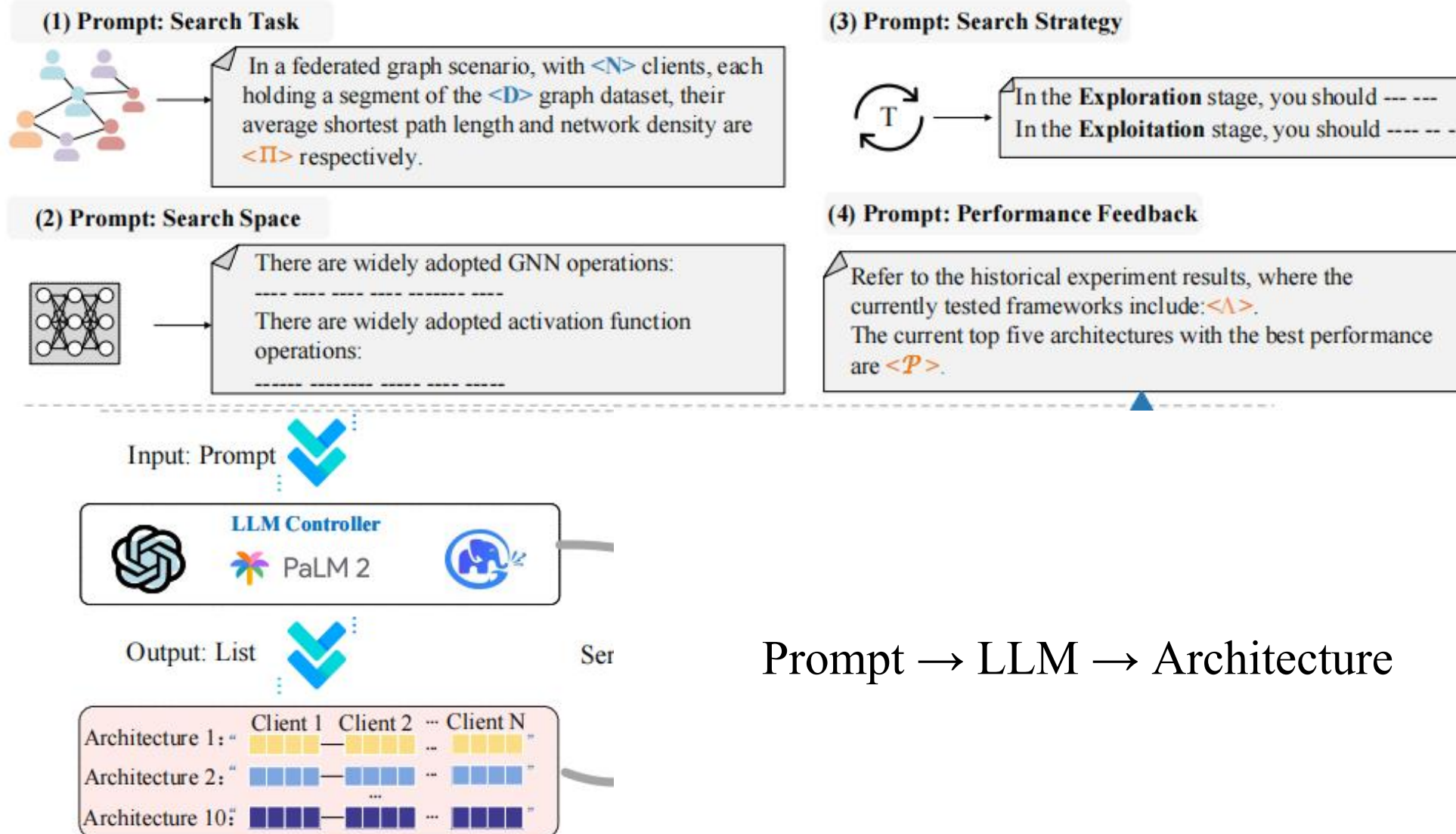
Deep Dive into PFGNAS

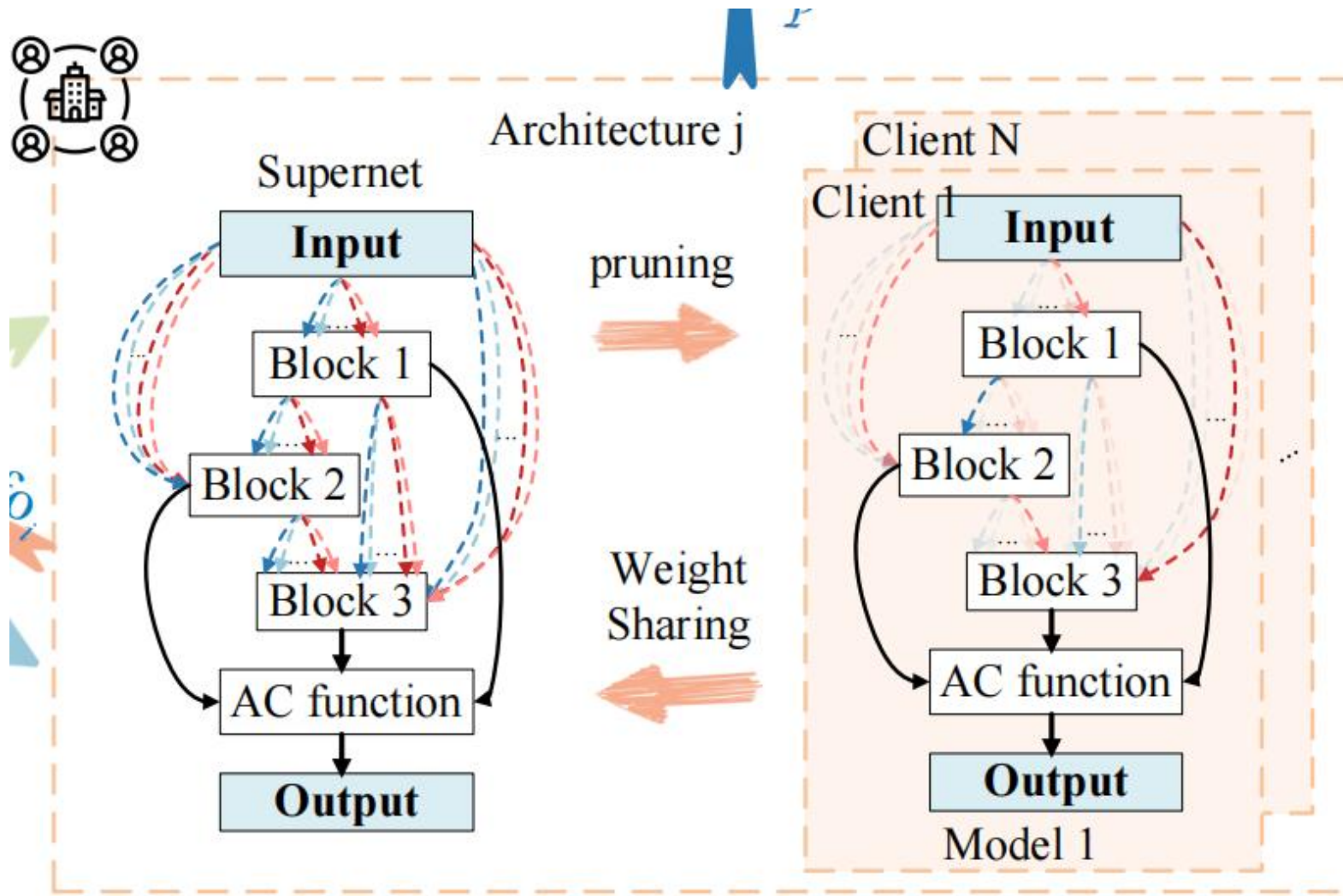
PFGNAS Overall Framework



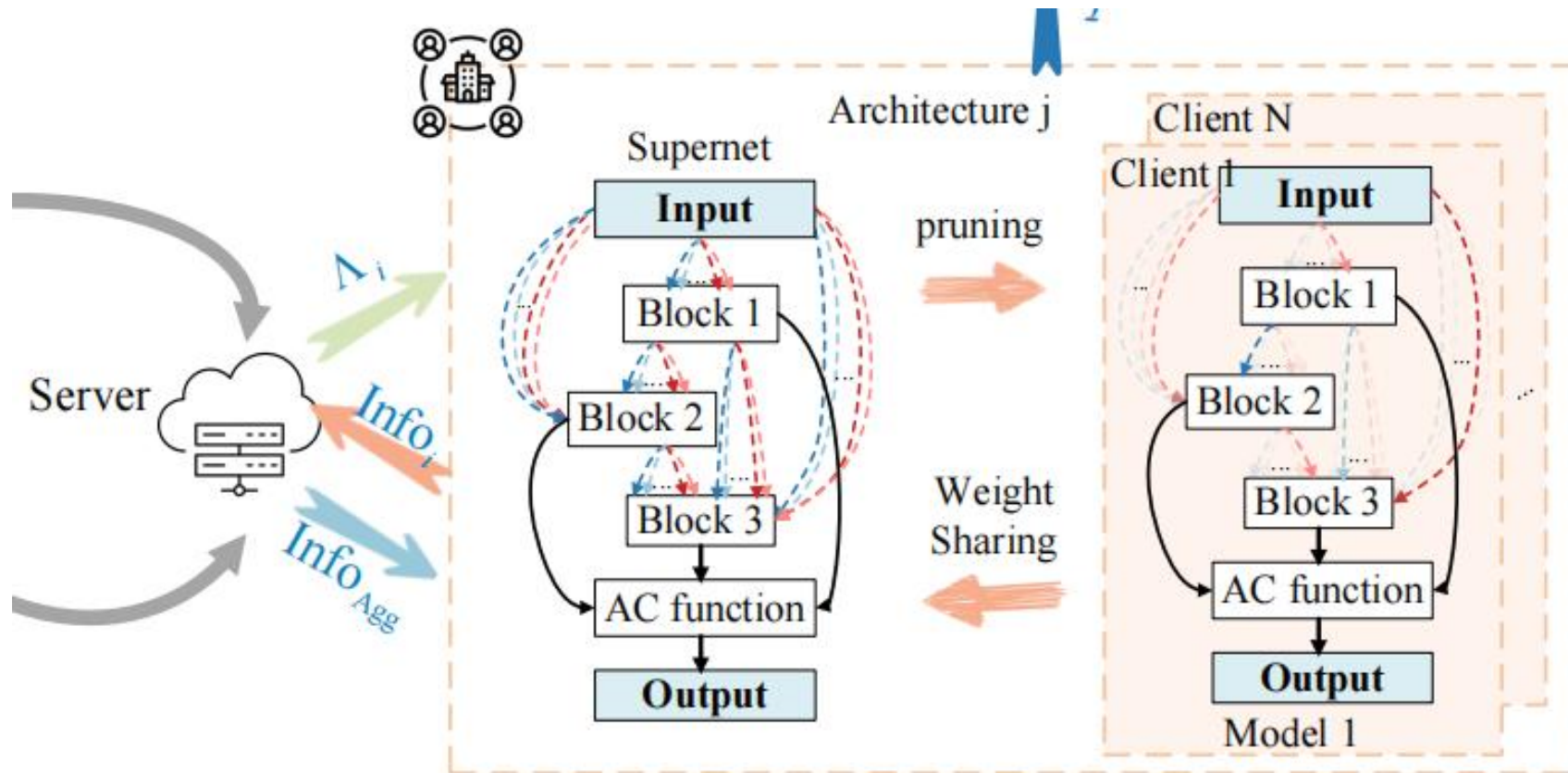
Communication Process:

1. Send architecture to Client
2. Send federated Info. to Server
3. Send aggregation federated Info. to Client
4. Send performance to Server





One network for all
architectures
Each architecture = a
subnetwork
Shared weight



- Server distributes architectures
- Clients train locally
- Updates are aggregated
- Feedback guides next round

Local Updates

Compute gradients only on pruned
local subnets

☒ Adapt to local topology; no
global parameter interference

Global Aggregation

Selectively aggregate activated
shared supernet params

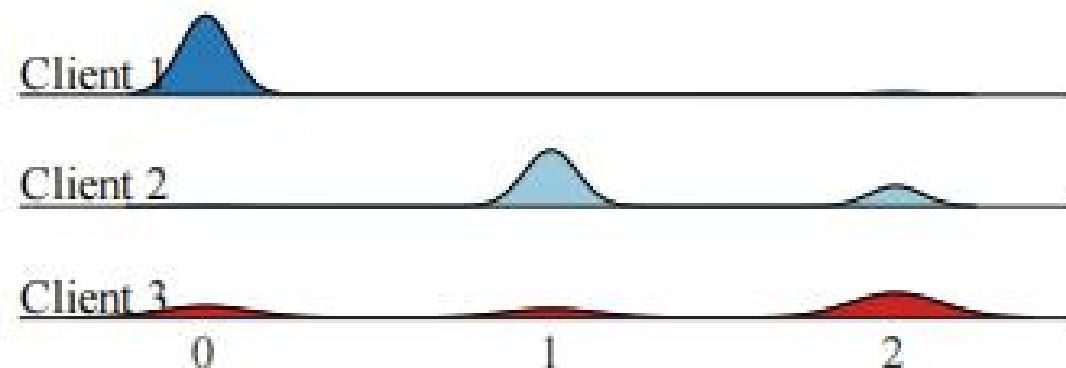
☒ Balance architectural
personalization & global
convergence

$$\Lambda^* = \operatorname{argmax}_{\Lambda \in \mathbf{LLM}(\Omega)} \mathcal{M}(\Lambda(G)), \quad (1)$$

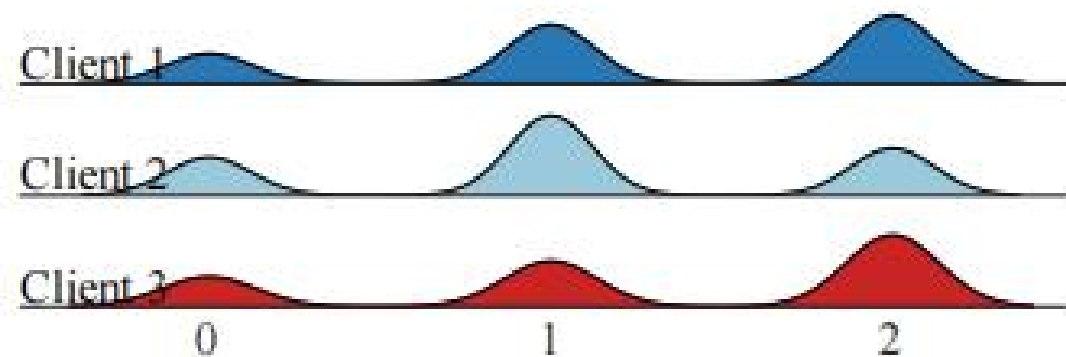
$$\min_{w \in \mathbb{R}^d} F(w) := \sum_{n=1}^N \frac{|G_n|}{|G|} f_n(w), \quad (2)$$

$$\begin{aligned} \Lambda^*(w^*) &:= \operatorname{argmax}_{\Lambda \in S} \mathcal{P}(\Lambda'(w^*), \Lambda(w^*)), \\ s.t. \ w^* &= \sum_{n=1}^N \frac{|G_n|}{|G|} E_{(G_i, y_i) \sim \xi_i} [\mathcal{L}(\theta_i; (G_i, \mathbf{Y}_i))], \end{aligned} \quad (3)$$

- Datasets: Cora, Citeseer, Pubmed
- Heterogeneity: Dirichlet distribution for data skew



(a) Pubmed ($N=3, \beta=0.2$)



(b) Pubmed ($N=3, \beta=10$)

Results: Outperforming Baselines

$\beta=10$

Model	Cora (3)	Cora (5)	Citeseer (3)	Citeseer(5)
GCN	78.01±0.42	76.90±0.85	78.96±0.54	<u>78.53±1.20</u>
GraphSage	75.70±0.42	77.66±1.19	77.56±0.95	<u>77.88±0.27</u>
GAT	76.22±0.91	76.13±1.51	79.92±0.80	78.63±1.24
GIN	68.97±1.28	66.84±0.87	59.94±2.76	64.75±2.97
SGC	76.30±0.53	76.05±1.39	80.20±0.40	78.10±0.31
ARMA	53.88±8.81	64.11±9.96	78.53±1.05	77.67±1.06
APPNP	74.08±2.52	70.25±5.61	<u>80.31±0.75</u>	77.67±2.12
FL+Darts	44.44±18.88	61.73±1.74	50.00±15.71	37.04±9.44
FL+Graphnas	27.16±1.74	37.04±4.00	51.11±1.81	56.29±4.19
FLAGNNS	76.19±1.63	82.20±1.27	74.29±1.54	77.15±2.10
FedPUB	<u>82.32±0.62</u>	80.65±0.08	73.17±0.33	73.71±0.59
PFGNAS	83.80±2.22	<u>80.86±1.73</u>	80.70±0.95	79.79±1.63

GLM-4	PFGNAS	84.20 _{±1.98}	-	80.09 _{±0.79}	-
GLM-4 Flash	PFGNAS	84.67 _{±1.34}	79.05 _{±2.02}	80.24 _{±1.47}	80.17 _{±0.27}

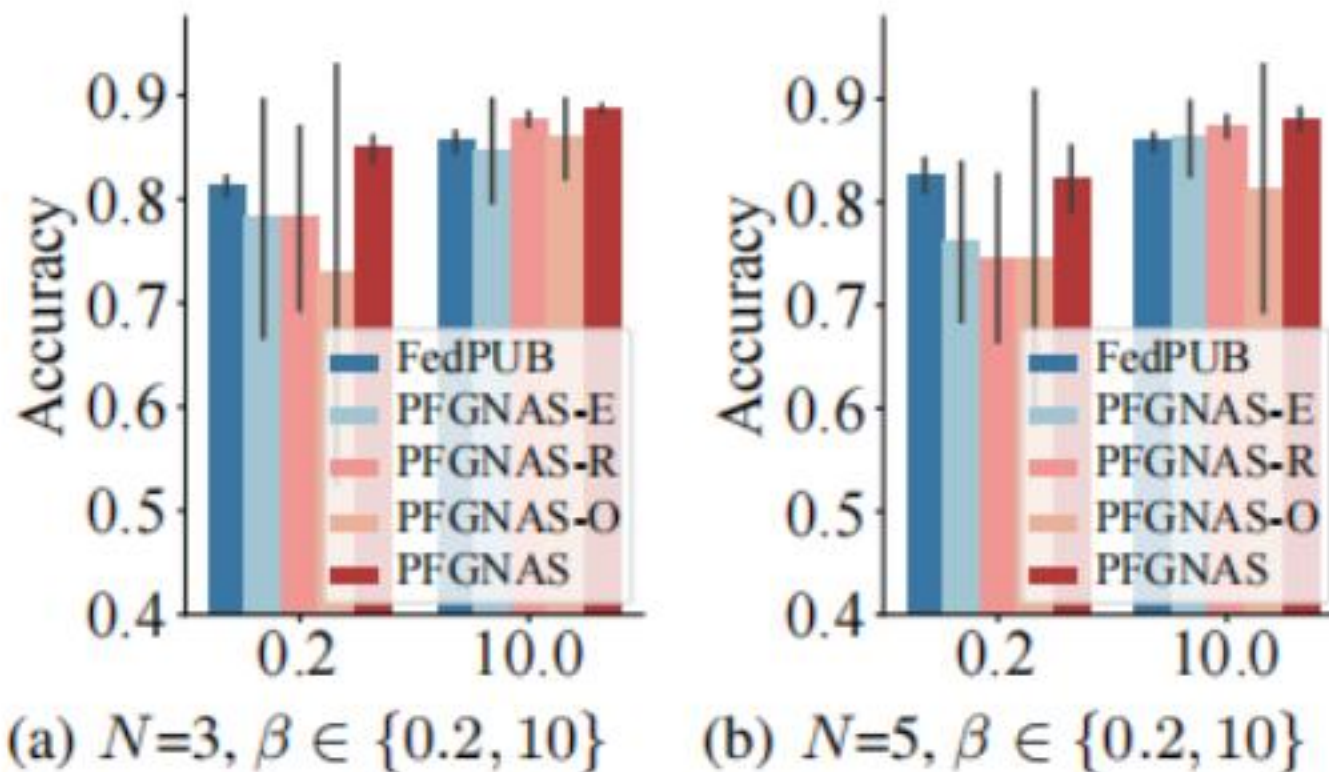


Figure 3: The global accuracy in different heterogeneous Pubmed data distribution. The black line located in the center of the bar chart signifies the error range.

$\beta=0.2$

Model	Client 1	Client 2	Client 3	Global	Avg $\uparrow$
FL+GCN	0.67	0.72	0.70	0.75	-
FedPUB	0.84	0.83	0.72	0.81	12.88%
FL+GraphNAS	0.00	0.48	0.67	0.77	-33.74%
PFGNAS-O	1.00	0.85	0.80	0.73	19.73%
PFGNAS-E	1.00	0.87	0.81	0.81	23.45%
PFGNAS	1.00	0.80	0.84	0.86	23.75%

Table 2: The accuracy of the PubMed dataset ($N=3$, $\beta=0.2$) on different clients, where Avg $\uparrow$ indicates the average improvement in performance compared to the baseline across all devices. The **bold** font highlights the best performance.

$\beta=10$

N=3

	Model	Accuracy	AUC
Citeseer	PFGNAS-GPT4	80.39 ± 2.05	79.46 ± 0.17
	PFGNAS-PALM	79.33 ± 0.73	78.36 ± 0.49
	PFGNAS-GLM4	80.70 ± 0.95	<u>79.32 ± 0.20</u>

Citeseer	PFGNAS-GLM4 flash	80.24 ± 1.47	78.85 ± 0.49
----------	-------------------	------------------	------------------

Pubmed	PFGNAS-GPT4	79.73 ± 4.18	93.79 ± 0.70
	PFGNAS-PALM	85.42 ± 3.62	94.75 ± 0.82
	PFGNAS-GLM4	86.52 ± 0.79	95.20 ± 0.84

Pubmed	PFGNAS-GLM4 flash	86.80 ± 0.83	95.32 ± 0.87
--------	-------------------	------------------	------------------

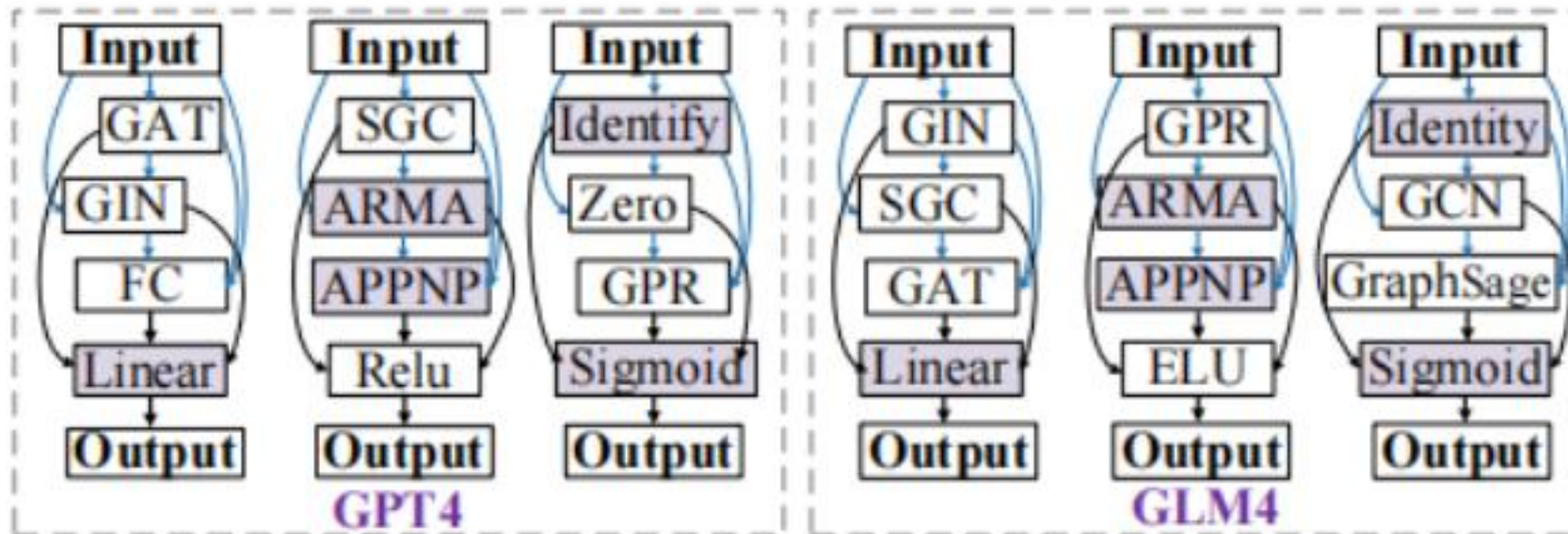


Figure 4: The comparison of the optimal architecture between GPT4 and GLM4 on Citeseer(N=3) dataset. Operators with the same insertion position and structure are highlighted with a purple background.

- 1、 Read the code carefully
- 2、 Reduce communication cost
- 3、 Introduce real-world heterogeneous graphs



Thanks



Algorithm 1: Personalized Federated Graph Neural Architecture Search (PFGNAS)

Input:

$\{G_i\}_{i=1}^N$: Graph datasets across N clients

Ω : Search space of GNN operations

LLM: Pre-trained large language model

T : Number of communication rounds

Output: Personalized architectures $\{\Lambda_i\}_{i=1}^N$

- 1: Initialize supernet S with shared weights W ▷ Global supernet
- 2: Initialize performance history $H \leftarrow \emptyset$ ▷ For feedback to LLM
- 3: **for** $t = 1$ **to** T **do** ▷ Communication round t
- 4: Construct prompt K_t consisting of:
- 5: – Search task (client number, dataset statistics)
- 6: – Search space Ω (candidate operations)
- 7: – Search strategy $\psi(t)$ $\psi(t) = \begin{cases} \text{exploration, if } t < T_{\text{threshold}} \\ \text{exploitation (self-reflection), otherwise} \end{cases}$
- 8: – Performance feedback H (historical results)
- 9: $\Lambda_t \leftarrow \text{LLM}(K_t)$ ▷ LLM generates candidate architectures

```
10:   for each client  $i \in \{1, \dots, N\}$  in parallel do                                ▷ Local training
11:       Select sub-architecture  $A_i$  from  $A_t$ 
12:       Extract corresponding subnetwork  $S_i$  from supernet  $S$ 
13:       Train  $S_i$  on local graph  $G_i$  to obtain weights  $W_i$ 
14:       Update the weights of  $S$  with  $W_i$                                           ▷ Weight sharing
15:   end for

16:   Aggregate/synchronize supernet weights  $W$                                 ▷ E.g., FedAvg
17:   Evaluate architectures on validation data and update history  $H$                 ▷ Feedback to LLM
18: end for
19: Return personalized architectures  $\{A_i\}_{i=1}^N$ 
```